

From Static Charts to Intelligent Learning Ecosystems: A Vision for the Future of ATC/ATM Training

Running Header: Future ATC/ATM Learning Ecosystems

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ABSTRACT

Air traffic control and management (ATC/ATM) training requires early-stage trainees to understand dense, spatially complex aviation information. However, foundational training continues to rely heavily on static charts and two-dimensional displays. **This paper presents a vision for the future of ATC/ATM training through immersive, adaptive, and collaborative learning ecosystems.** Grounded in several years of iterative prototype development and feedback from aviation subject matter experts and certified professional controllers, the work explores how browser-based XR, dynamic FAA data integration, embodied interaction, and collaborative learning environments can support spatial understanding and operational learning. **Building upon this foundation, we envision future ATC/ATM training ecosystems centered around three complementary educational modalities: self-paced learning, AI-supported scenario-based practice, and instructor-led collaborative learning.** Over the next 5–25 years, these ecosystems may expand access to training, strengthen self-efficacy, support workforce development, and enable more personalized and experiential pathways for expertise development. More broadly, ATC/ATM serves as a high-complexity testbed for investigating how immersive technologies, human-centered AI, and collaborative learning infrastructures can be integrated to support professional learning in safety-critical domains. This work contributes a future-oriented HCI perspective on the design of immersive learning ecosystems for complex operational environments.

CCS CONCEPTS • Human-centered computing → Virtual reality • Human-centered computing → Collaborative and social computing • Applied computing → Interactive learning environments

Additional Keywords and Phrases: Air Traffic Control and Management, Extended Reality, Immersive Learning Ecosystems, Human-Centered AI

1. INTRODUCTION

Air traffic control and management (ATC/ATM) remains a safety-critical occupation with exceptionally high training demands. Yet, the workforce pipeline continues to face mounting pressure from staffing shortages, retirements, and trainee attrition. **The Federal Aviation Administration (FAA) recently reached a 27-year low in controller staffing, with nearly one-third of controllers projected to become retirement-eligible by 2030 and substantial workforce losses anticipated throughout the 2021–2030 period [6].** Simultaneously, early-stage trainees often experience low self-efficacy while learning airspace concepts through traditional 2D Visual Flight Rules (VFR) maps, a process widely characterized as tedious, cognitively demanding, and difficult for novice trainees to navigate [17,22]. **As a result, trainees must rapidly internalize dense spatial and procedural information within instructional environments that still rely heavily on static memorization-based learning approaches [17].** These challenges have intensified the need for scalable, human-centered training systems that improve engagement, spatial understanding, knowledge retention, and workforce readiness. Emerging immersive technologies offer a promising pathway toward modernizing ATC/ATM training by transforming how complex airspace information is learned, interpreted, and collaboratively understood. Over the next 5–25 years, immersive ATC/ATM training systems may evolve from supplemental visualization tools into persistent, AI-supported collaborative learning ecosystems that reshape workforce development, procedural learning, and operational preparedness.

Among emerging immersive technologies, eXtended Reality (XR) has gained increasing attention as a medium for supporting self-paced learning, collaborative instruction, and more accessible operational training experiences. Existing research across safety-critical and cognitively demanding domains, together with insights from certified professional controllers (CPCs) and immersive experience experts, suggests that VR is a viable medium for ATC map visualization and

training with the potential to improve engagement, motivation, spatial understanding, and knowledge retention [8,10,12,23]. These environments may be further enhanced through AI-guided learning modules that support adaptive instruction, personalized feedback, and individualized learning experiences. VR training environments are particularly valuable because they enable repeated practice without the risks, costs, or operational limitations associated with real-world training scenarios [5,14].

Despite these advantages, immersive ATC/ATM training systems remain in the early stages of development and integration within safety-critical operational environments. Current research and applied training frameworks show limited integration of XR technologies for scalable instruction, collaborative training, and adaptive educational support. In response, this work outlines a long-term vision for ATC/ATM training that moves beyond static visualization tools and toward integrated learning ecosystems capable of supporting self-paced study, AI-mediated instructional guidance, and one-to-many instructor-led collaborative learning. Building upon this foundation, the paper envisions future integration with FAA Next Generation Air Transportation System (NextGen) technologies, training facilities, and educational systems to support more accessible, adaptive, and distributed workforce development over time.

While this vision is grounded in ATC/ATM training, the challenges it addresses are not unique to aviation. ATC/ATM is one instance of a broader class of professional-learning problems characterized by dense spatial and procedural information, high-stakes evaluation that makes real-world practice risky or costly, and expertise traditionally developed through instructor-guided apprenticeship. Domains as varied as surgical and emergency medicine, disaster response, cybersecurity operations, and industrial process control share these properties. We therefore treat ATC/ATM not only as an application area but as a high-complexity testbed for a general HCI question: how can immersive, adaptive, and collaborative learning ecosystems be designed to support expertise development in safety-critical, spatially demanding professions? We return to this framing in Section 6, where we articulate the properties that determine when the model presented here is likely to transfer.

2. BACKGROUND

2.1. Embodied Spatial Cognition and Operational Understanding

Embodied spatial cognition provides an important theoretical foundation for immersive ATC/ATM training because air traffic control learning depends heavily on interpreting complex spatial relationships, procedural environments, and operationally relevant airspace information. By treating understanding as a product of physical perception and interaction rather than abstract symbolic processing alone, immersive systems may reduce cognitive load while supporting spatial interpretation, situational awareness, and long-term knowledge retention [15,20,21]. Embodied learning research suggests that immersive environments can strengthen comprehension when bodily action, spatial interaction, and task-relevant representations are meaningfully aligned with the concept being learned [3,16,18].

Prior work on 3D visualization supports this theoretical direction. Bogomolova [13] explored the role of 3D visualization in anatomy learning and found that students who used 2D images often needed to infer 3D anatomical structures from limited spatial information. In contrast, immersive 3D visualization presents these relationships directly, reducing the burden of mental reconstruction while making spatial context easier to interpret. Similar challenges exist in ATC/ATM training, where trainees must interpret dense symbolic chart information while simultaneously developing operational awareness. This requires contextual understanding of complex airspace structures and spatial relationships.

Feedback from SMEs and CPCs further supports this direction by suggesting that immersive map visualization can make VFR sectional information feel more contextual, interactive, and spatially coherent than traditional 2D study methods [17,22]. These findings suggest that immersive spatial environments may support not only map interpretation but also broader operational understanding by allowing trainees to experience airspace relationships in ways that more closely resemble real-world navigation and decision-making contexts.

2.2. Self-Efficacy, Persistence, and Early Training Experience

Bandura describes perceived self-efficacy as learners' task-specific beliefs about their capability to organize and execute the actions needed to achieve desired outcomes, rather than a generalized sense of confidence or self-worth [2]. Multon, Brown, and Lent further found that self-efficacy beliefs are positively associated with both academic performance and academic persistence [11]. Within ATC/ATM training environments, where learners must quickly develop proficiency in complex spatial and procedural tasks, self-efficacy may play an important role in persistence, engagement, and long-term training success.

XR may support this process by enabling repeated low-risk procedural rehearsal before high-pressure operational evaluation. These immersive environments provide psychologically safer and more repeatable opportunities to practice, explore, and refine difficult tasks before entering operational training contexts. Empirical work has shown that VR-based

instruction can improve self-efficacy [4]. Evidence from aviation, emergency response, and medical training further supports the value of self-efficacy within immersive learning environments [4,7,19].

Despite these promising findings, longitudinal research examining XR-supported self-efficacy development in safety-critical operational domains such as ATC/ATM remains limited. This gap represents an important opportunity for future immersive training research because self-efficacy may influence academic persistence, long-term trainee retention, and broader workforce development outcomes. Taken together, these findings suggest that XR training platforms may support not only academic performance and persistence but also broader workforce development by enabling repeated low-risk practice, adaptive learning opportunities, and more accessible pathways for early operational skill development.

2.3. Human-Centered Challenges for Aviation XR Systems

Although XR offers significant promise for aviation training, safety-critical immersive systems must be designed around human needs, cognitive limitations, and operational demands rather than technical novelty alone. Within ATC/ATM training environments, higher realism is not automatically beneficial. Immersive systems must balance representational fidelity, usability, instructional clarity, and cognitive workload so that learners can focus on operationally relevant tasks without becoming overwhelmed by interface complexity or visual clutter.

High presence and visual realism may increase engagement and motivation; however, they can also introduce extraneous cognitive load when immersive interface elements compete with procedural understanding, spatial interpretation, and decision-making demands [8]. These challenges are particularly important in ATC/ATM, where attentional management, situational awareness, and information prioritization are essential for effective learning and operational performance. Adaptive interaction strategies that allow learners to control informational complexity may help reduce visual clutter and support progressive exposure to information as self-efficacy develops over time.

Interface usability remains another important consideration. Ongoing involvement from CPCs and SMEs can help ensure that immersive training interfaces align with operational expectations and real-world procedural reasoning. As these systems evolve, AI-guided instructional support may further assist learners by prioritizing operationally relevant information based on contextual learning needs. Feedback from SMEs and CPCs also reinforces the importance of guided instruction and classroom integration, suggesting that future ATC/ATM XR systems should carefully balance immersive realism with cognitive workload management, adaptive support, and instructor-led guidance [17].

2.4. WebXR as Infrastructure for Scalable Access

To support a scalable and continuously evolving ATC/ATM training ecosystem, the underlying architecture must support maintainable content updates, operational data integration, and distributed immersive access. WebXR serves as a foundational infrastructure by enabling reusable workflows for integrating evolving FAA airspace information while reducing the need to manually reconstruct training environments following chart updates.

By supporting browser-based deployment and synchronized multi-user environments, WebXR expands access to immersive training while preserving embodied VR interaction across devices. These capabilities support self-paced learning, instructor-guided collaboration, and geographically distributed participation, making WebXR a practical foundation for future ATC/ATM learning ecosystems.

3. FOUNDATIONS OF THE FUTURE ECOSYSTEM

The future learning ecosystem envisioned in this work is grounded in several years of iterative, stakeholder-driven development. Early phases of the project engaged aviation SMEs, CPCs, and immersive technology experts to identify critical challenges within existing ATC/ATM training workflows. These discussions consistently highlighted VFR map memorization as a major obstacle during early training and identified immersive VR as a promising modality for supporting spatial understanding, engagement, and experiential learning [22].

These findings informed the development of an immersive prototype focused on layered airspace visualization and self-paced exploration. Successive usability evaluations with novice users and CPCs demonstrated strong usability (SUS = 73.2–85) [1,9], perceived training value, and opportunities for enhanced realism, instructional guidance, and instructor-facing functionality [17,22]. These recommendations directly informed the current generation of the platform, which migrated to a WebXR framework to support scalability, dynamic FAA data integration, and collaborative learning. Additional enhancements include controller-based map manipulation, integrated 3D airspace structures, and synchronized interactions across linked sessions, establishing the technical foundation for future instructor-guided learning environments.

4. NEAR-TERM VALIDATION PATHWAY

Building upon the developmental foundation established through prior prototype iterations, the immediate focus of this work is empirical validation. The first step is to evaluate whether immersive 3D airspace visualization provides measurable advantages over traditional 2D instructional methods for learning VFR sectional information. Controlled studies will assess spatial understanding, map memorization, knowledge retention, and learner self-efficacy. Consistent with prior educational XR research, these measures will be operationalized using validated instruments for spatial understanding, knowledge retention, and self-efficacy rather than ad hoc assessments, allowing results to be interpreted against established findings in immersive science and anatomy learning [10,13,23]. Critically, efficacy measurement is positioned at the earliest stage of the validation pathway rather than deferred until later ecosystem features are built, so that higher-order capabilities are grounded in demonstrated learning benefits rather than assumed ones. Establishing the educational value of immersive 3D representations is a necessary prerequisite for future ecosystem development.

Following this initial validation, the platform will be deployed as a supplemental educational resource to support self-paced exploration and independent study. This phase will examine learner engagement, adoption patterns, and self-efficacy development while refining embedded assessment and learning analytics capabilities. Positive outcomes would justify broader instructional integration.

The next stage focuses on classroom deployment and longitudinal evaluation. By integrating the platform into active educational environments, researchers and instructors can examine how immersive learning experiences influence self-efficacy, persistence, knowledge retention, and training outcomes over time. Together, these efforts provide the empirical foundation necessary for advancing toward larger-scale learning ecosystems that integrate collaborative instruction, adaptive learning support, and AI-guided educational experiences.

5. FUTURE WORK & IMPACT

Over the next 5–25 years, advances in AI, immersive technologies, and distributed learning infrastructures may enable ATC/ATM training to evolve beyond static memorization-based approaches toward adaptive, collaborative, and experiential learning ecosystems centered around self-paced learning, scenario-based practice, and instructor-led collaboration.

5.1. Self-Paced Learning Ecosystems

Future ATC/ATM learning environments may expand opportunities for self-directed study outside traditional classroom settings through browser-based deployment and immersive visualization. AI-supported learning modules could personalize training by identifying strengths and weaknesses, recommending targeted practice activities, and adapting instructional content to learner progress. Knowledge tracing algorithms may further support competency-based progression, providing learners with greater agency while creating opportunities for repeated low-risk practice and self-efficacy development.

5.2. Scenario-Based Learning Ecosystems

Scenario-based learning represents one of the most promising opportunities for future immersive ATC/ATM education. Rather than relying primarily on memorization and procedural instruction, trainees could engage with realistic operational situations that require active problem-solving, decision-making, and application of learned concepts. AI-assisted scenario generation may support a wide range of training experiences, from routine traffic management operations to emergency events and weather-related disruptions, while allowing instructors to dynamically adjust scenario complexity and conditions to meet instructional objectives. Such capabilities would enable repeated exposure to operational situations that are difficult to recreate consistently within traditional training environments.

Beyond procedural learning, scenario-based ecosystems may help trainees develop confidence, stress-management strategies, and operational judgment through repeated practice in increasingly complex situations. This experiential learning model more closely reflects the realities of ATC/ATM operations and may better prepare trainees for future workplace demands.

5.3 Instructor-Led Collaborative Learning Ecosystems

Although AI and self-paced learning may expand access to training opportunities, instructors will remain a critical component of future learning ecosystems. Future collaborative environments may support one-to-many instruction within synchronized immersive spaces where instructors demonstrate operational techniques, discuss decision-making strategies, and guide trainees through realistic scenarios. This shifts instruction from teaching procedures toward facilitating operational understanding through shared experiences.

AI-assisted analytics may further support instructors by identifying common trainee errors, highlighting learners who require additional support, and summarizing performance trends across a cohort, helping direct instructional attention where human feedback and mentorship provide the greatest educational value.

5.4 Workforce and Educational Transformation

The long-term impact of immersive learning ecosystems may extend beyond individual training outcomes to influence workforce development across the broader ATC/ATM pipeline. By expanding access to training and supporting repeated practice, immersive systems may help learners build confidence while improving learning efficiency, knowledge retention, and preparedness for advanced stages of training.

6. BROADER HCI IMPACT: ATC/ATM AS A MODEL FOR IMMERSIVE LEARNING ECOSYSTEMS

Although grounded in ATC/ATM training, the contribution of this work is a transferable model rather than an aviation-specific result. ATC/ATM is best understood as one instance of a broader class of safety-critical, spatially demanding professional-learning problems, and the design directions developed here raise questions that extend across the HCI community. In the subsections that follow, we articulate when this model is likely to generalize and surface a set of open questions intended to invite cross-domain discussion. Each subsection concludes with a question we hope will engage workshop participants working well beyond aviation.

6.1 A Transfer Framework: When Does This Model Generalize?

Rather than asserting that ATC/ATM simply resembles other domains, we propose a small set of properties that determine when an immersive learning ecosystem of the kind described here is likely to transfer. We suggest a domain is a strong candidate when it exhibits: (1) high spatial and procedural complexity that requires learners to reconstruct three-dimensional structure from two-dimensional representations; (2) a safety-critical context in which real-world practice is risky, costly, or difficult to reproduce; (3) reliance on situational awareness and information prioritization across a complex operational display; (4) expertise developed through instructor-led apprenticeship combined with repeated rehearsal and self-efficacy development; and (5) a need for scenario variety that is difficult to stage consistently in reality, favoring AI-assisted generation.

These criteria make the model's reach concrete. Surgical and anatomy education satisfies criteria (1), (2), and (4), where trainees must mentally reconstruct spatial structure and rehearse before high-stakes performance [13]. Emergency and disaster response satisfies (1), (3), and (5), demanding real-time spatial coordination under conditions that are hazardous to reproduce. Cybersecurity operations and industrial control rooms satisfy (3) and (4), where practitioners maintain situational awareness across dense operational displays and develop expertise through mentored practice. Viewed through these criteria, ATC/ATM is not a special case but a representative one, which is precisely what makes it a useful testbed.

Discussion prompt: Which of these properties are necessary versus merely sufficient for transfer, and what safety-critical domains does the model fail to capture?

6.2 Human-Centered AI and Learner Agency

Current AI-supported learning research has not established how instructional authority should be distributed among the learner, instructor, and AI system, particularly in collaborative immersive environments. Educational AI commonly emphasizes prediction, personalization, and individual tutoring, while instructors and learners have limited control over when assistance is provided, how learner activity is interpreted, and whether recommendations are accepted (Celik et al., 2022; Alfredo et al., 2024). These questions become more complex in immersive learning because an AI system can interpret selections, movement, response times, errors, and other embodied actions. Such observations may support instruction, but they may also produce inaccurate conclusions about learner understanding or reduce learners' opportunities to make independent decisions.

The longer-term ATC/ATM platform could provide a testbed for examining how different allocations of AI authority affect learner agency, instructor decision-making, and independent learning. Studies could compare direct AI guidance, instructor-mediated recommendations, learner-requested assistance, and conditions without AI support. The platform could also test whether instructor-reviewed scenario generation supports useful practice without introducing operational errors or excessive dependence on automated guidance (Ghaffari et al., 2025). Outcomes should include perceived control, reliance on AI, instructor workload, unaided performance, retention, and transfer to unfamiliar tasks. This would extend teacher-AI complementarity research into asymmetric immersive instruction while examining when AI supports learning and when it begins to displace learner reasoning or instructor authority

Discussion prompt: How can AI-mediated training augment rather than displace human instructors while preserving learner agency in safety-critical domains?

6.3 Collaboration and Social Presence at a Distance

Prior research shows that asymmetric VR can support collaboration between immersed learners and non-immersive partners, but it also identifies problems with workspace awareness and divided attention. In a VR/tablet pair-learning study, asymmetric and symmetric conditions produced comparable learning outcomes, although the symmetric condition provided stronger presence, immersion, and awareness of the partner's activity. The same study found that salient visual signaling helped direct learners' attention (Drey et al., 2022). Similar systems have allowed instructors to monitor multiple learners, switch between their first-person views, and manipulate the virtual environment from a tablet or desktop (Wang et al., 2023). However, these systems have received limited classroom evaluation. More broadly, a systematic review identified only 11 empirical studies of collaborative immersive learning and emphasized the need for better support for joint attention, spatial awareness, and marking shared objects (Paulsen et al., 2024).

These limitations motivate the longer-term development of the present system from one-to-one guidance toward one-to-many ATC instruction. Synchronized highlighting, layer states, task instructions, and event overlays would provide shared instructional references across the desktop and immersive interfaces. The instructor interface would also need to present each learner's current view, progress, and requests for assistance without imposing excessive monitoring demands. Future studies should examine whether these mechanisms help instructors establish joint attention, maintain awareness of individual learners, and provide timely guidance across multiple headsets. This work would address a question that remains insufficiently studied: what information and controls must be shared across asymmetric interfaces for one instructor to guide several immersive learners effectively?

Discussion prompt: What forms of co-presence and shared spatial grounding are required for effective one-to-many instruction in immersive environments?

6.4 Assessment and Evidence of Learning in Immersive Ecosystems

Assessment remains a central challenge in immersive learning research because usability, presence, engagement, and confidence are often reported without establishing whether learners retain knowledge or transfer it beyond VR. Reviews of safety-relevant training show that few studies evaluate subsequent behavior, long-term outcomes, or transfer to real tasks (Stefan et al. 2023), while experimental evidence demonstrates that stronger presence does not necessarily produce better learning (Makransky et al., 2019). The present project should therefore evaluate educational efficacy before treating its adaptive, collaborative, or AI-supported capabilities as beneficial. Comparisons between immersive visualization and paper charts should combine immediate and delayed knowledge tests with transfer tasks using unfamiliar chart content. Accuracy, errors, response time, and understanding of airspace relationships should serve as primary outcomes, with usability, cognitive load, self-efficacy, and interaction logs used as explanatory measures. Because recorded behavior may reflect interface proficiency rather than aviation knowledge, system telemetry should not be interpreted as competence without external validation. This raises a broader HCI question: what evidence is required to show that an immersive learning ecosystem supports durable and transferable learning rather than only a favorable experience?

Discussion prompt: What should count as evidence of learning in immersive ecosystems, and how early must efficacy be demonstrated before building on unvalidated assumptions?

6.5 Access, Equity, and Scalable Infrastructure

The WebXR foundation described in Section 2.4 reflects a deliberate bet on browser-based deployment as a path to scalable, distributed training. This choice lowers barriers to access, but it also raises equity questions that the HCI community is well positioned to examine: disparities in device capability, network access, and institutional resources may shape who benefits from immersive professional training. Designing infrastructure that broadens access without reproducing existing inequities is a challenge shared by any domain pursuing distributed workforce development.

Discussion prompt: How can browser-based immersive infrastructure broaden access to professional training without reproducing existing inequities?

7. CONCLUSION

Modern ATC/ATM operations require trainees to understand dynamic three-dimensional airspace, yet early training continues to rely heavily on static charts and two-dimensional displays, creating a mismatch between how airspace is represented during training and how it must be understood in practice. By representing airspace information as interactive 3D environments, immersive technologies offer a promising pathway toward supporting spatial understanding through exploration, interaction, and embodied learning. The future opportunity extends beyond visualization. Learning ecosystems that combine self-paced study, scenario-based practice, and instructor-led collaboration may expand access to training, strengthen self-efficacy, and support workforce development. Supported by adaptive instructional systems, scalable WebXR infrastructures, and AI-assisted learning tools, these ecosystems may fundamentally shift training from facility-bound memorization toward continuous, learner-centered educational experiences that better prepare future professionals for increasingly complex operational environments. By integrating embodied interaction, human-centered AI, collaborative instruction, and scalable learning infrastructures within a unified educational ecosystem, this work contributes to a broader HCI vision for the future of professional learning, workforce development, and expertise preparation in complex operational domains.

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